

An Approach of Large-scale Multi-agent Simulation Using Combat Modeling

Toshihiko Yamamoto^a, Akira Namatame^a, Hiroshi Sato^a

^aDepartment of Computer Science, National Defense Academy

1-10-20 Hashirimizu Yokosuka 239-8686, Japan

g47038@nda.ac.jp

Experience in a wide range of fields for which complex adaptive systems exist has shown that increasing the detail of the model does not necessarily improve its ability to reproduce reality, but often simply reflects the emphasis placed on various aspects by the designer. This complexity makes modeling of a combat situation difficult, particularly because the model designer must assume knowledge of how the participants will behave in a given situation. Thus there is a real danger that the model will simply reproduce the preconceived ideas of the modeler.

With the growing acceptance of agent-based models as a method for simulating combat, which are designed to incorporate behavior which adapts to the situation, it may be that a clearer link between fractal statistics and the non-linear dynamics of combat can be established. Combat must generally be considered to be a complex adaptive system that evolves in an unpredictable way. Complex networks in many fields have been shown to generate discontinuous functions to describe various quantities as functions of space and time, which often show scaling similarities. The modern combat simulation be modeled with simplified agent-based models combined with complex networks, in which an agent's behavior is determined by some set of rules, which are also constrained by the networks of friends and enemies.

War game simulations are often used for analysis and training. However, traditional linear models of combat situation such as Lanchester equations treat attrition as a continuous function of time. However, recent thinking on the nature of warfare has it that combat is an inherently nonlinear situation. Attrition is during duels followed by agent movements in space and time. Tactics will try to concentrate duels for all individuals. If the statistical moments of the data scale, it can be expected that the variance relative to the size of the force will increase as the size of the force decreases. The dependence of the statistical variance on the size of the force results from using an attrition rate which is not continuous.

The interesting question is then, what happens to the statistics if the opposing forces are allowed to position themselves in a dynamically evolving and non-linear way? Lauren[1] shows that such functions are well described by fractal models. He investigated the possibility of using a fractal to describe attrition is examined using Cellular automaton. Essentially, the existence of such scaling means that combat has structure on many scales, so that small-scale combat resembles large-scale combat. At each scale there is a battle and a campaign equivalent. At low levels, terminology tends to engagement and battle. At the lowest level, there are duels and series of duels. The existence of such a self-similar hierarchy provides phenomenological justification for modeling combat in terms of aggregated units, a common practice for war games. If combat outcomes for large units may be treated as an aggregation of the outcomes of combat between sub-units occurring on shorter characteristic time scales, the statistics of the outcomes are likely to be scale dependent.

This means that, for example, the variance of the casualty ratio increases as the size of the units involved in the combat decreases. Lauren examined the possibility that combat statistics are scaling, and hence have fractal indices. He suggests that the statistical properties of combat may be described in terms of fractal dimensions. The main consequence of this kind of statistical behavior is that the combat should be more variable for smaller-scale forces. This has significant implications for the statistical understanding of combat simulation, particularly modeling combat using the statistics of scaling systems, providing justification for modeling combat in terms of aggregated units, and suggesting that there exists some basic law of combat behavior. Given the high degree of variability of small-scale combat, it seems that modeling such combat explicitly for analysis purposes may be of limited value unless large numbers of simulations are run. Also, historical data suggests that reasonably fat-tailed distributions with greater extremities than normal distributions must be produced by a model if it is to accurately represent reality.

He hypothesizes that the scaling-law is valid in real combat mean that the statistics should show behavior typical of multi-scaling fractal (multi-fractal) fields, such as fat-tailed probability distributions. This has significant implications for the risk associated with war-fighting. Unfortunately, this hypothesis is difficult to prove without detailed data of real conflicts which is unlikely to become available.

If small-scale combat has a much higher degree of variance, as suggested here, explicit models of combat on this scale may often be of limited use for analytical purposes. At the very least, it must be expected that such simulations must be run hundreds of times in order to gain knowledge of the distribution of outcomes. On the other hand, large-scale combat outcomes are much less variable, suggesting that simulations of combat between large forces may be usefully predictive. In this case, it might be just as useful for analytical purposes to use simple combat models which describe the evolution of the combat in terms of generalized events, provided these models can reproduce the appropriate statistics. Such simple models may be easily run a large number of times, thus providing a more complete picture of the statistical variations likely for small-scale scenarios.

Keywords

agent-based simulation, flocking behavior, combat analysis, 3D-simulation, lanchester equation

References

[1] Lauren. M, "Fractal Methods Applied to Describe Cellular Automaton Combat Models," Fractal, Vol.9, No.2, 2002.